Complete figure book: clinical course, serum markers and EASI estimation

This review copy contains the five main figures followed by all 21 supplementary figures. Captions specify populations, uncertainty and what each comparison can establish. Complete numerical results are supplied as CSV source tables.

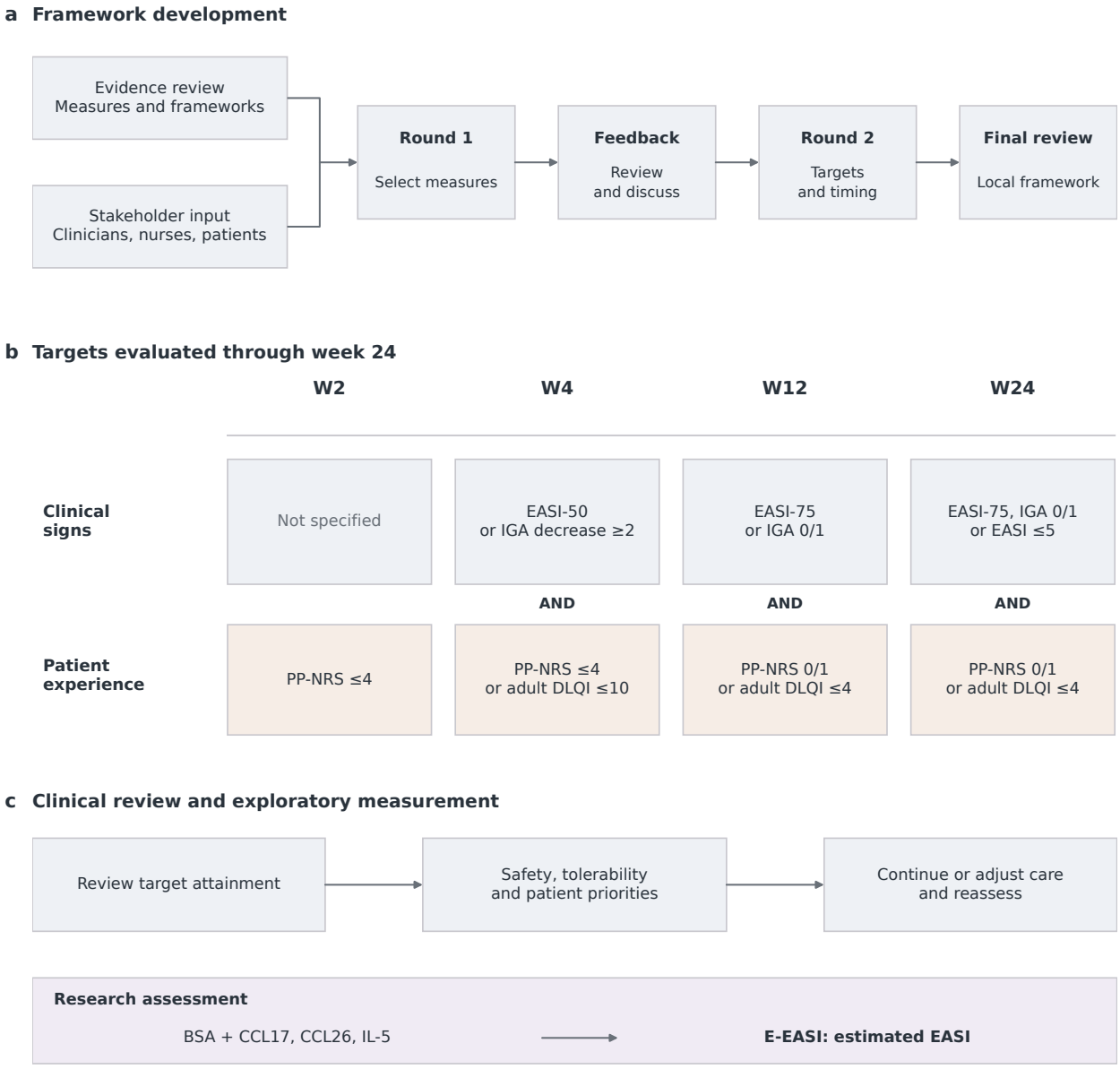


Figure 1: **Development, clinical targets and the E-EASI research assessment.** a, The project materials describe evidence review, clinician/nurse/patient input, two questionnaire rounds, feedback and steering-group review. The planned agreement threshold was 75%; 49 returned questionnaires were reported. Original ballots and round-specific participation and agreement are unavailable, so the diagram reconstructs the reported process rather than verifying consensus. b, Operational targets for W2/W4/W12/W24. At W2 only PP-NRS $\leq$4 is required. From W4 onward at least one criterion in each domain is required; "or" is within a domain and "AND" joins domains. EASI-50/75 means at least 50%/75% improvement from positive pretreatment EASI; IGA decrease $\geq$2 is relative to baseline. IGA 0/1 and PP-NRS 0/1 mean recorded values 0 or 1. DLQI is analysed in adults aged $\geq$18. Missing components remain unknown unless another observed component resolves the domain. The recorded IGA instrument and implemented framework version require investigator confirmation. Longer-term planned targets and other published frameworks are compared in Table S13. c, Target review accompanies safety and patient priorities. E-EASI is a separate, post hoc hybrid clinical–molecular research estimate; it did not determine treatment changes. This observational cohort does not establish that the framework or the research estimate improves management.

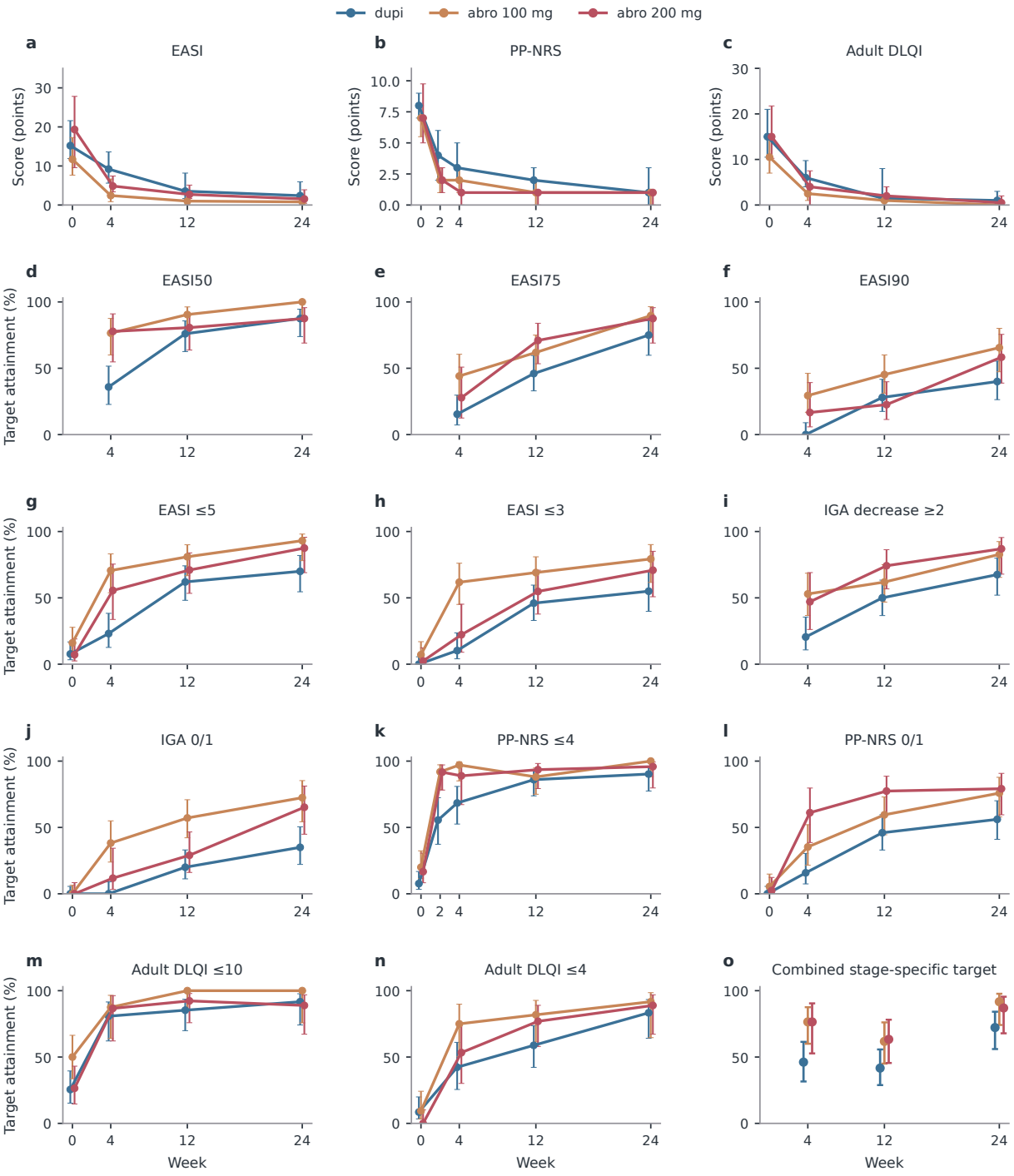


Figure 2: **Observed clinical activity and target attainment by initial treatment.** a–c, Observed medians and interquartile ranges for EASI, PP-NRS and adult DLQI. d–n, Clinical threshold attainment, with pointwise Wilson 95% confidence intervals. Each visit uses its available observations, not a fixed patient set. Group positions are slightly offset for legibility. Improvement thresholds require observed baseline and follow-up; adult DLQI requires age ≥18. o, Combined physician-and-patient targets from Fig. 1, with Wilson intervals. These stage-specific criteria differ, so points are not connected. Colour denotes initial treatment throughout; subsequent treatment changes do not reassign groups. No outcomes are imputed. All target numerators and denominators are in Supplementary Table S1; headline missing-outcome bounds are in Results and headline_missing_outcome_bounds.csv. Baseline-adjusted treatment comparisons are in the supplement. dupi denotes dupilumab; abro denotes abrocitinib. These available-case, observational trajectories do not establish causal treatment differences.

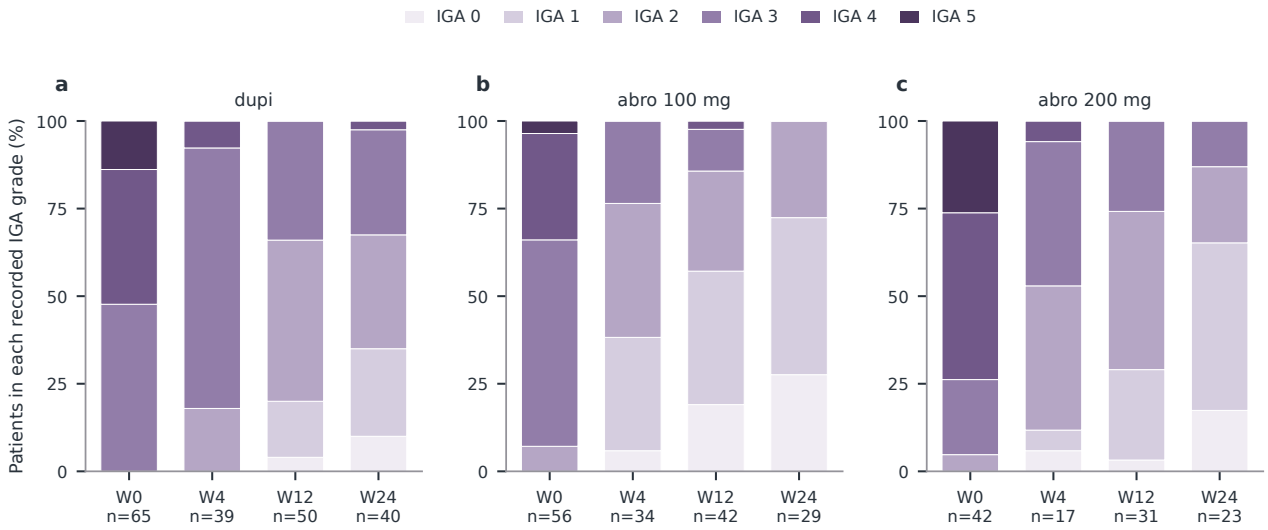


Figure 3: **Recorded IGA distributions during follow-up.** a–c, Distribution of recorded IGA grades within each initial treatment group at W0, W4, W12 and W24. Each bar totals 100% of the patients with an observed grade; n below the bar is that visit's denominator. Colour identifies grade, not treatment. Patient sets may differ between visits. Grade-specific counts and proportions are provided in the source table. The local 0–5 coding is retained; correspondence to a validated IGA instrument has not been confirmed. This figure describes disease activity, not dose-reduction success.

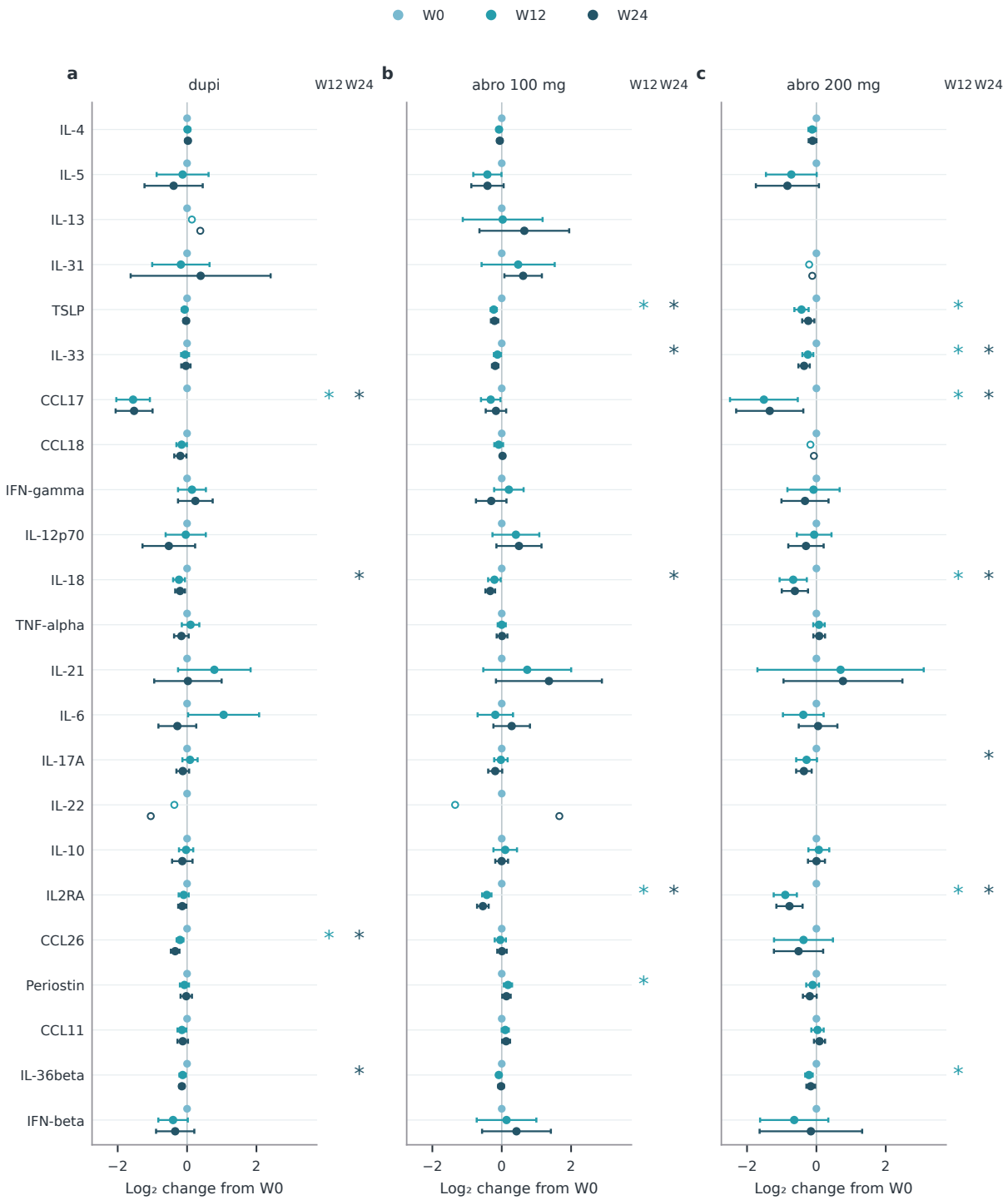


Figure 4: **Molecular changes across three visits in the same patients.** a–c, All 23 serum markers by initial treatment. For each marker and treatment, the same patients contribute at W0, W12 and W24; patient sets can differ between markers and treatments. W0 is the reference value zero, not zero concentration. Coloured points show mean within-person $\log_2$ changes from W0; horizontal bars are 95% t confidence intervals for n≥5. A change of −1 means a geometric paired concentration ratio of 0.5. Hollow later-visit points denote n<5 without inference; blank rows have no complete three-visit set. The two star columns labelled W12 and W24 refer to the corresponding visit versus W0: *BH q<0.05 across all 138 planned marker-by-treatment-by-contrast tests, using paired sign-flip tests and retaining non-estimable hypotheses in the family. These stars are calculated in the displayed fixed patient sets, not copied from analyses of all available pairs. No baseline covariate adjustment is applied to this descriptive figure. Between-treatment adjusted comparisons and all available-pair windows are reported separately in the supplement. Baseline concentrations are given in Table S12; exact n, changes, intervals and q values are in the source table.

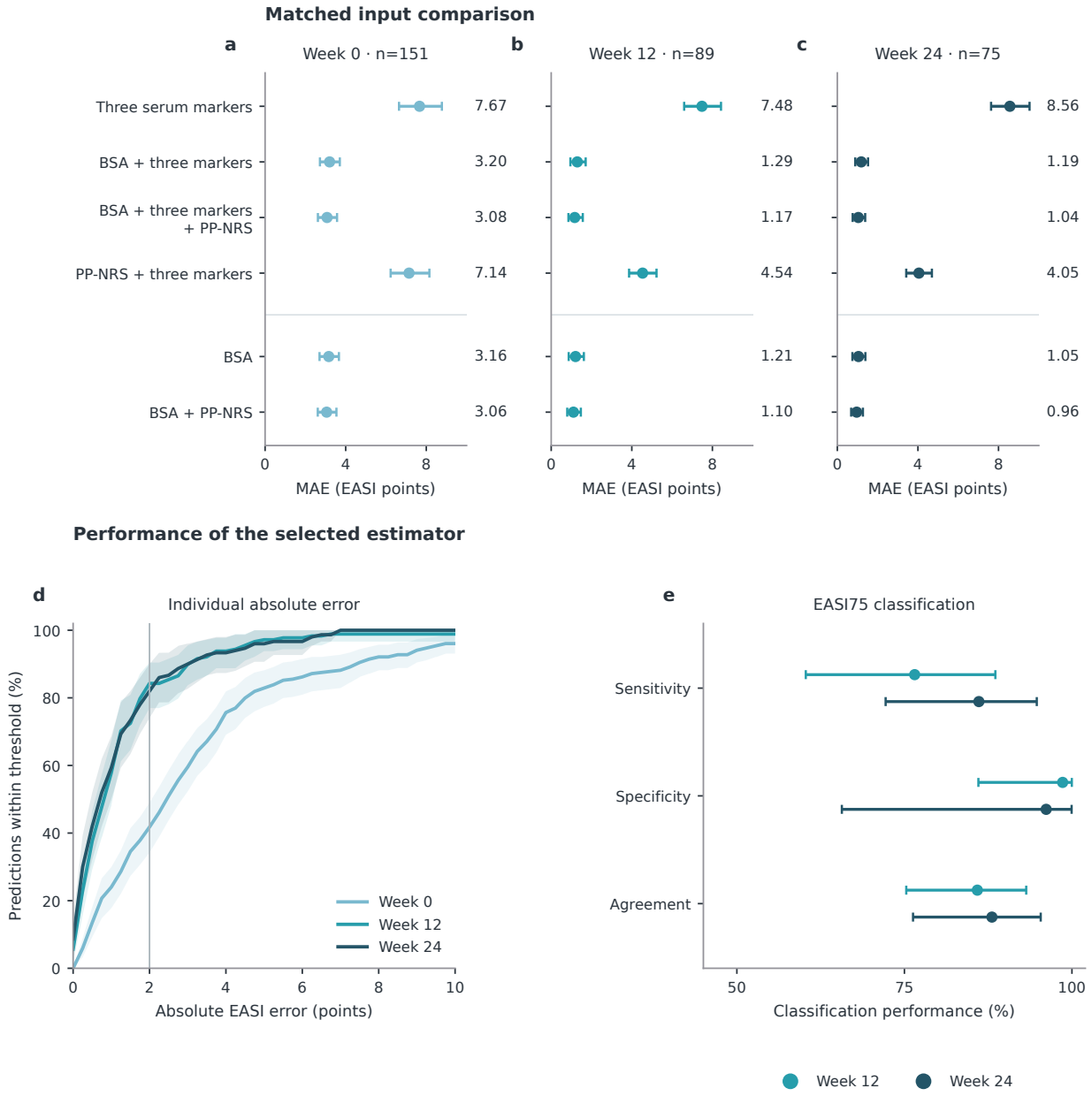


Figure 5: **Input contributions and performance of the common EASI estimator.** a–c, Post hoc input-set comparisons on the same 151/89/75 patients at W0/W12/W24. Points and labels show mean absolute error (MAE) with 95% paired-patient bootstrap intervals. Markers are CCL17, CCL26 and IL-5; clinical inputs come from the current visit. Models use the same outer patient folds, W0/W12 training, and training-only preprocessing and inner selection. Clinical-only controls appear below the divider; molecular added accuracy requires comparison at identical clinical inputs. d, The unchanged primary procedure on its original 152/89/75 patients: cumulative absolute-error proportions with pointwise bootstrap 95% intervals. W0/W12 predictions are patient-held-out. Only one W12 patient and no W24 patients have observed EASI >21. All errors enter the calculation; thresholds up to ten points are displayed. Two points is a visual reference, not an accepted interchangeability margin. e, EASI75 classification uses observed and estimated scores with their own baselines. Sensitivity uses 49/54 responders, specificity 36/13 nonresponders and agreement 85/67 paired patients at W12/W24. Conservative conditional 95% bounds average repetition-specific Clopper–Pearson limits with Bonferroni adjustment across repetitions within each endpoint, visit and metric, not all displayed metrics. The binomial working model permits dependent repetitions but does not propagate shared fitted models or development uncertainty. Metrics are scored per repetition before averaging; intervals condition on saved predictions. W24 is previously inspected same-cohort application. Separate Wilson intervals and counts are in Supplementary Table S11; additional input comparisons, CCL17 controls and marker contributions are in Tables S6–S7.

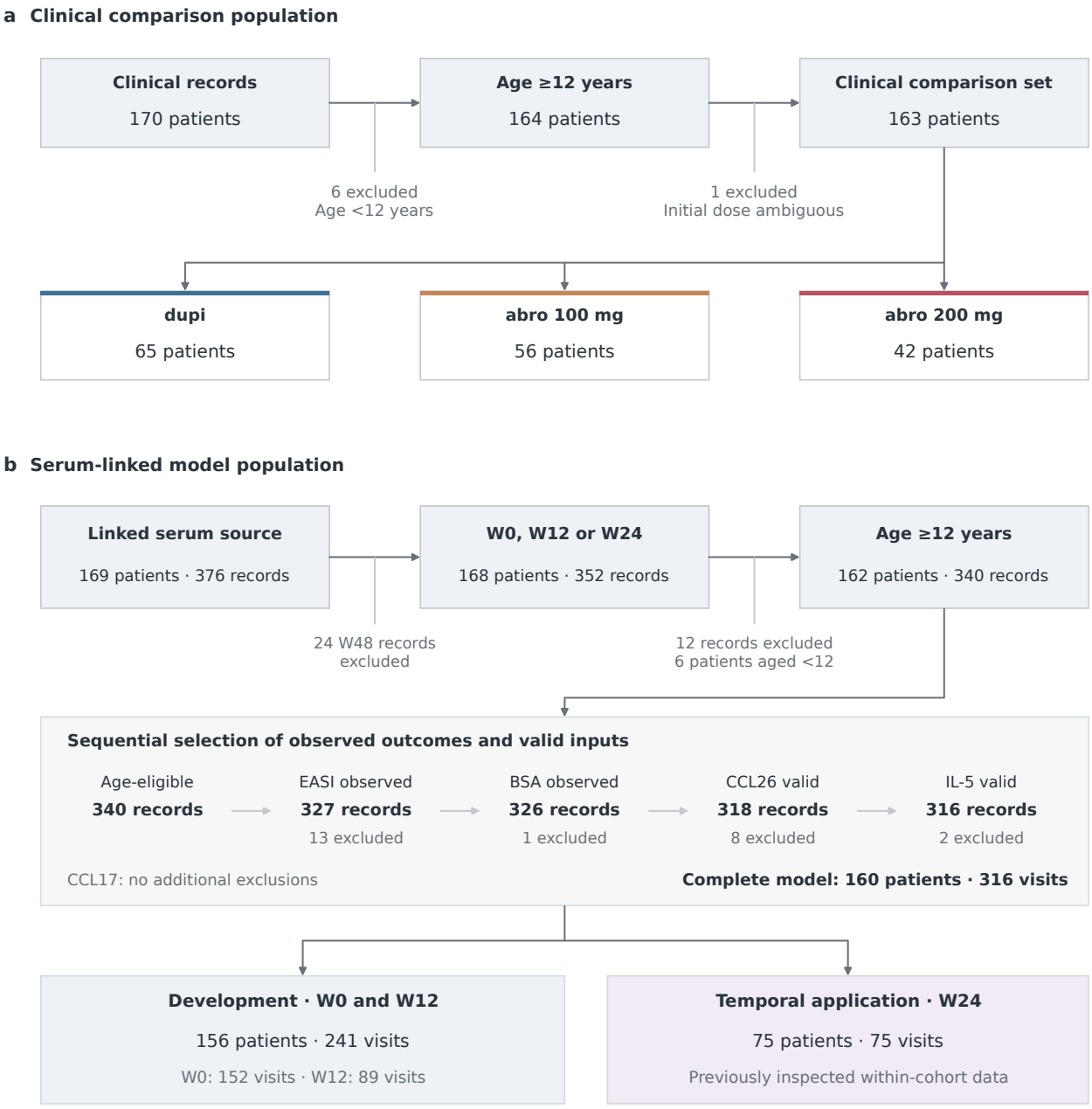


Figure S1: **Patient and record flow into the clinical and model analyses.** a, The curated clinical source contains 170 patients, not a screened or consented recruitment denominator. Six patients aged <12 years and one age-eligible patient with an ambiguous initial dose are excluded, leaving 163 patients classified by initial treatment: dupi, 65; abro 100 mg, 56; and abro 200 mg, 42. b, The linked serum source is a separate entry branch; model eligibility is not conditional on membership of the clinical comparison population. Removing 24 W48 records leaves 168 patients with 352 W0/W12/W24 records; age exclusions remove 12 records from six patients. Further ordered exclusions are mutually exclusive at the record level: 13 EASI, one BSA, zero CCL17, eight CCL26 and two IL-5 inputs are unavailable, leaving 316 complete model visits from 160 patients. A valid molecular input requires an available positive concentration for the log2 transformation. Loss of a record does not necessarily remove a patient. Development uses 241 W0/W12 visits from 156 patients, with patient-level validation splits; W24 includes 75 visits from 75 patients and is previously inspected within-cohort temporal application, not external validation. Development and W24 share 71 patients, with 85 development-only and four W24-only patients. Fifty-one patients have all three complete visits. The clinical and complete-input model populations share 159 patients, with four clinical-only and one model-only patient; the treatment-free model can retain an ambiguous initial dose. Scheduled empty visit slots are not counted as observed visits, and no filled-in clinical outcome is used as validation truth.

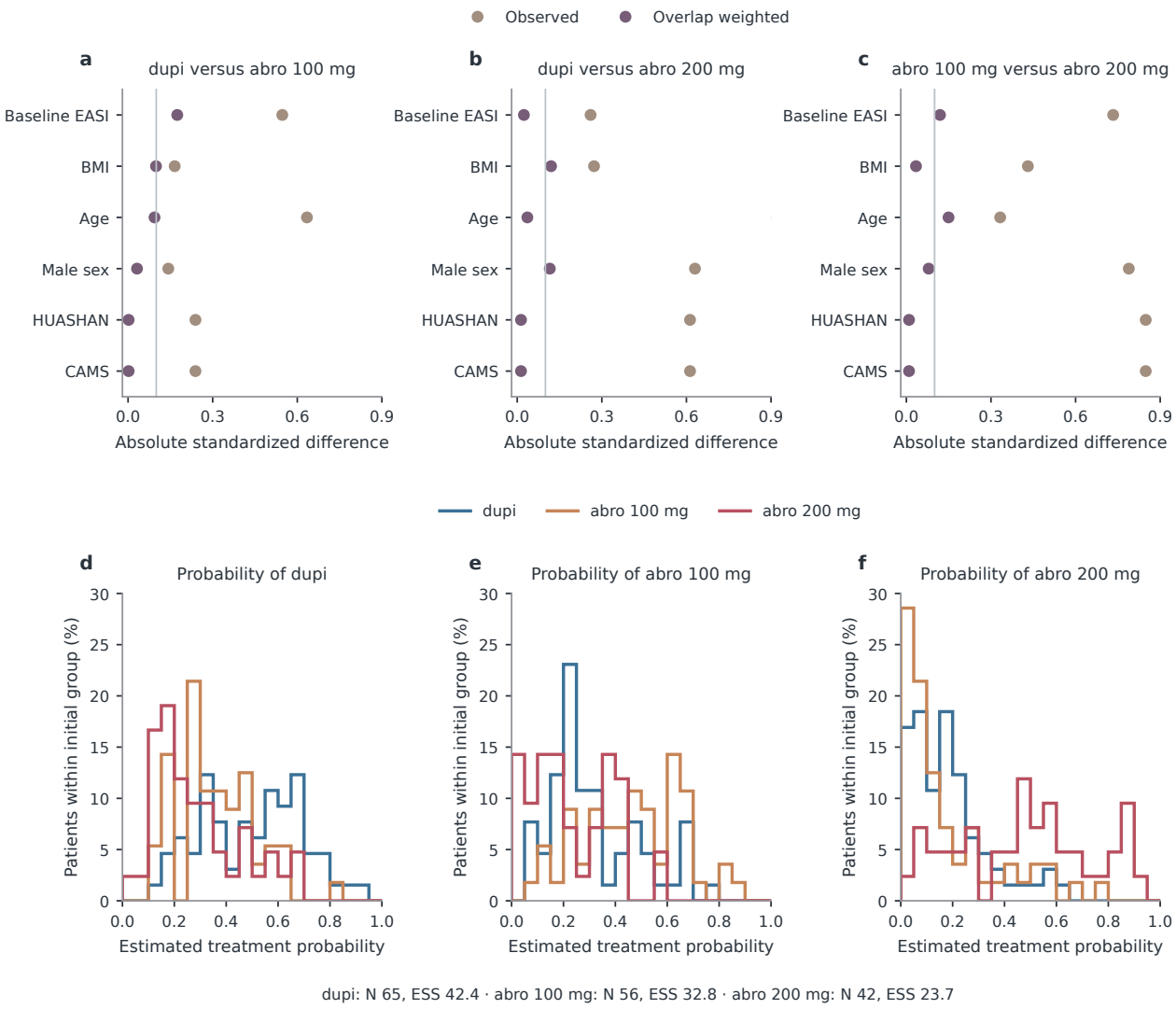


Figure S2: **Baseline balance, treatment overlap and effective sample size.** a–c, Absolute standardized mean differences for every covariate in the initial-treatment propensity model, before and after generalized overlap weighting. Differences use the unweighted pairwise pooled standard deviation in both displays. The light 0.10 reference line is a descriptive balance guide, not a statistical significance threshold. Residual imbalance remains. d–f, Baseline multinomial treatment probabilities, stratified by actual initial treatment; each distribution uses all patients in that initial group and sums to 100%. These show common support rather than longitudinal drug exposure. The footer reports original N and effective sample size, $(\text{sum w})^2/\text{sum}(\text{w}^2)$. All centre, sex, age, BMI and EASI variables are shown. Overlap weighting targets comparable baseline patients; it cannot remove unmeasured confounding or correct outcome-driven dose changes. Endpoint- and marker-specific availability/ESS diagnostics are retained in the source tables.

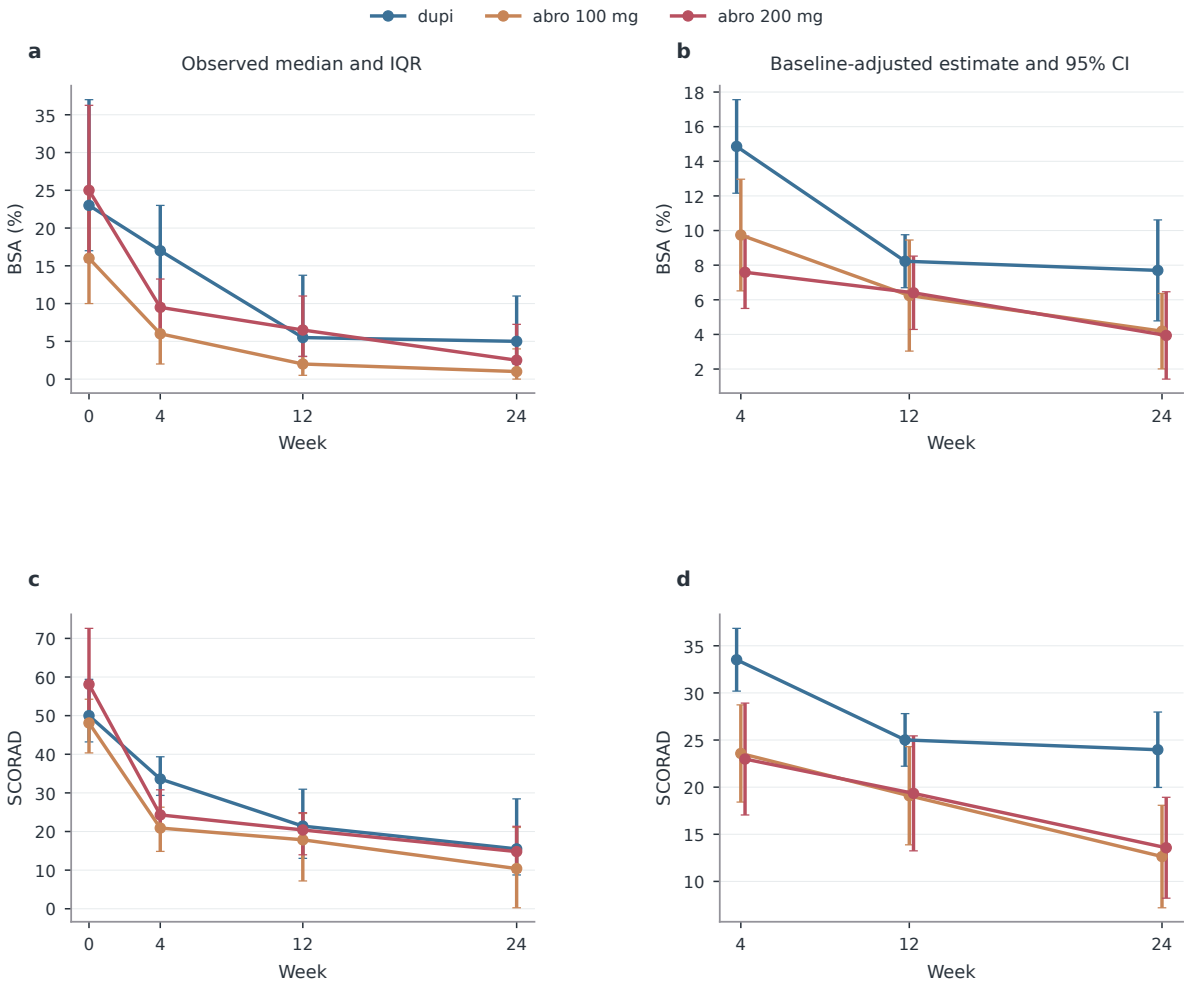


Figure S3: **Observed and adjusted BSA (%), SCORAD.** Left, observed medians with interquartile ranges by initial treatment; all available values are used at each specified visit. Right, generalized-overlap-weighted marginal estimates with patient-cluster robust 95% confidence intervals conditional on the estimated baseline weights, adjusted for baseline outcome, age, sex, BMI and centre. Patient sets can differ between visits. Exact outcome-specific counts and numerical comparisons are in the source tables; they are not replaced by a common denominator. Initial treatment groups describe observational management strategies, including subsequent regimen changes, rather than randomized fixed-dose effects.

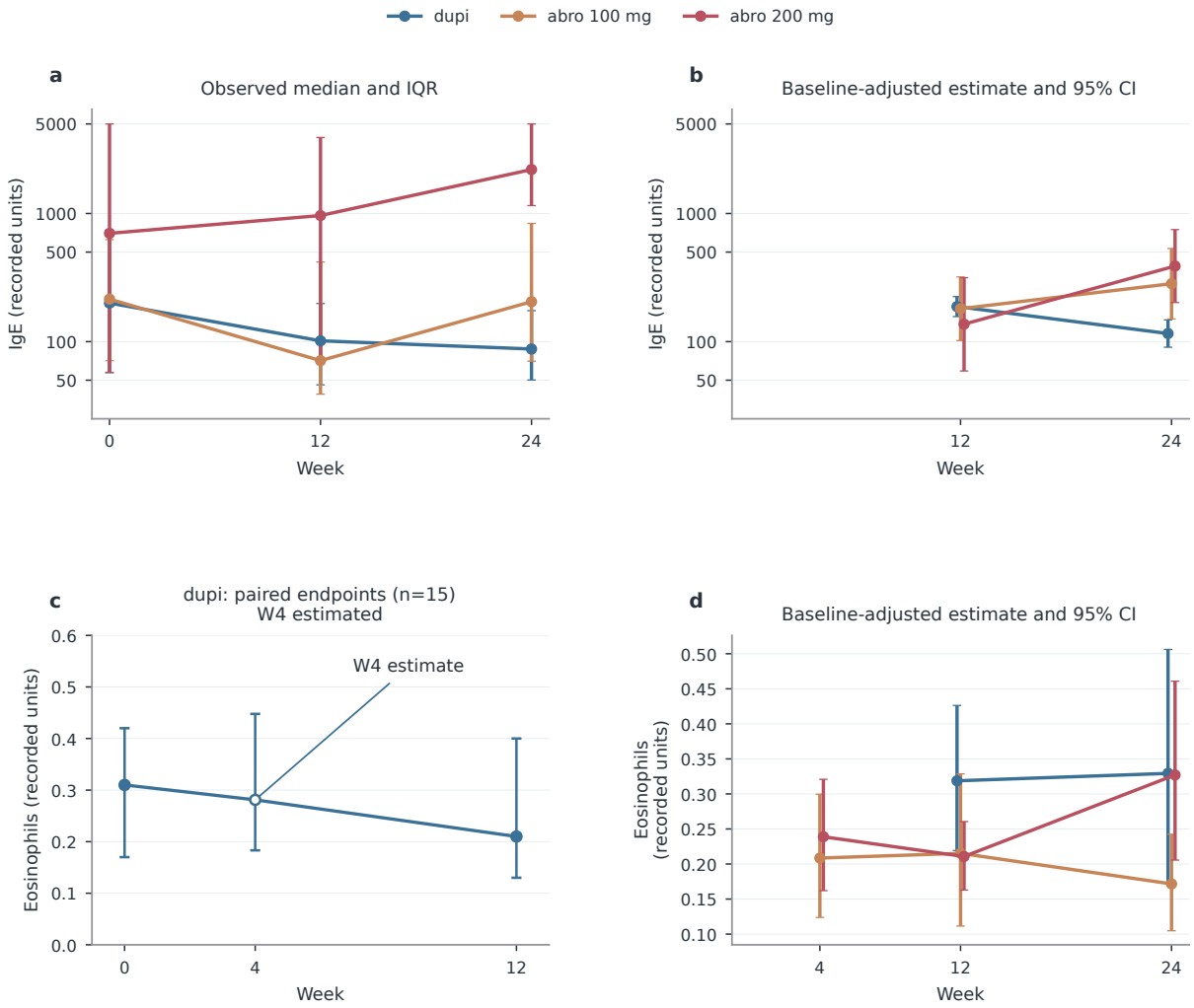


Figure S4: **IgE follow-up and the unobserved W4 eosinophil value.** a, Available-case IgE medians and interquartile ranges; patient sets vary by visit. The vertical axis is logarithmic with labels in recorded units. b,d, Generalized-overlap-weighted estimates from log(1+x) models adjusted for baseline outcome, age, sex, BMI and centre, with patient-cluster 95% confidence intervals conditional on estimated weights. Back-transformed estimates are not arithmetic means. There are no observed dupi W4 eosinophil measurements, so d contains no estimate for that treatment/visit cell. c, A separate fixed set of 15 dupi patients with W0 and W12 eosinophil measurements: filled points are observed endpoint medians; the hollow W4 point is an assumption-based estimate. Each patient's log(1+x) eosinophil value is linearly interpolated one-third of the way from W0 to W12, back-transformed, then summarized. Bars are patient-bootstrap 95% intervals for the median, conditional on this temporal assumption. The W4 estimate is 0.281 (0.183–0.448); it is not an observed result, validation outcome or treatment-comparison input. Original-scale interpolation, alternative time shapes and a seven-patient complete three-visit analysis are supplied separately. Initial treatment groups retain subsequent regimen changes.

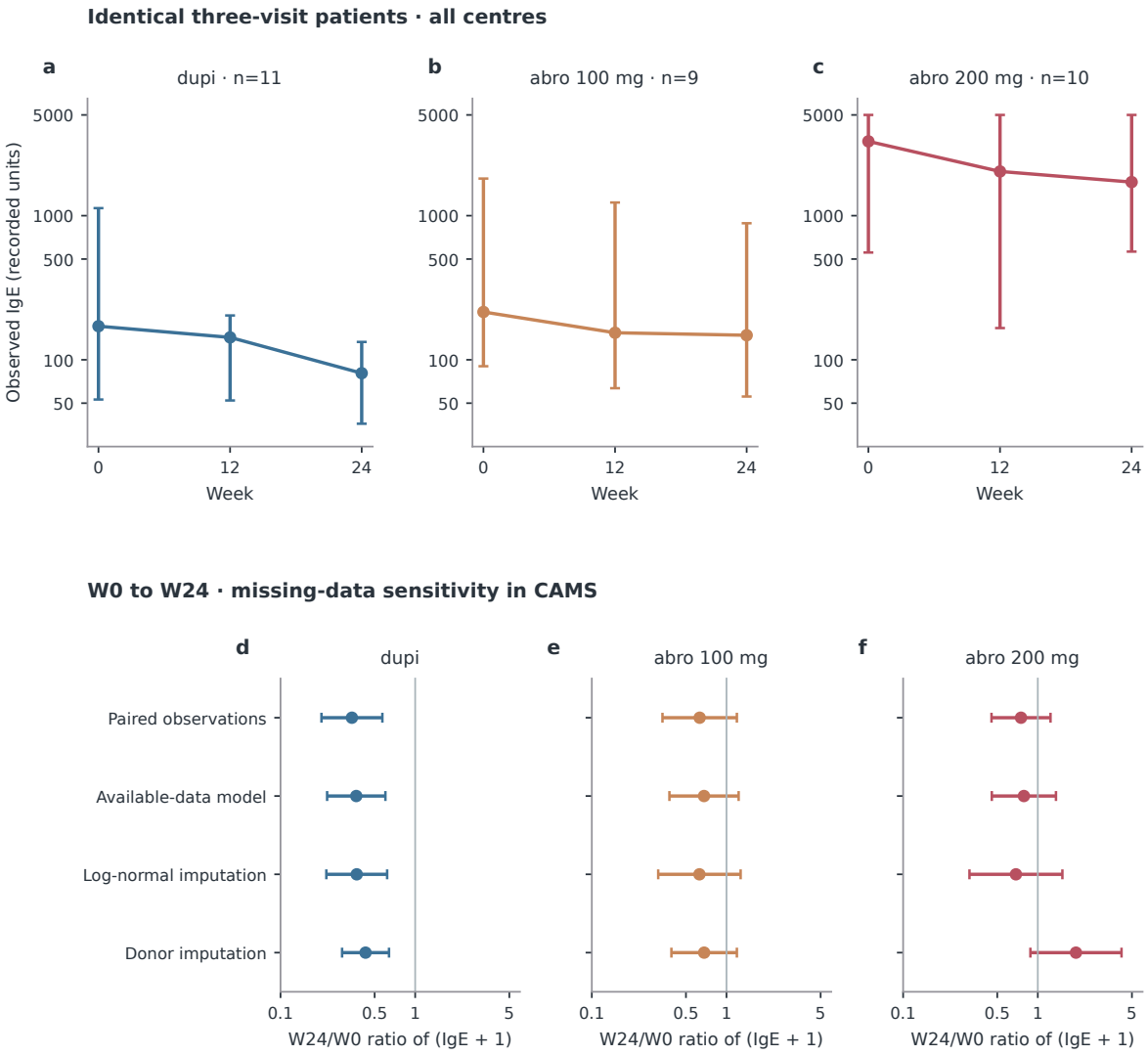


Figure S5: **IgE: fixed-patient observations and missing-data sensitivity.** a–c, Observed medians and patient-bootstrap 95% confidence intervals in the same complete three-visit patients at every visit: 11 dupi, 9 abro 100 mg and 10 abro 200 mg patients across both centres. The vertical scale is logarithmic, labelled in recorded units. d–f, W0–W24 geometric mean ratios of IgE+1 from four analyses within CAMS. The observed paired analysis uses all available W0/W24 pairs (14/10/12 by treatment); it does not require W12. The available-data random-intercept model uses 229 measured visits from 125 patients across the three treatments. Multiple-imputation analyses target all 130 CAMS patients (51/38/41), using patient-wide chained equations with either posterior log-normal draws or within-treatment predictive mean matching (donor imputation); 400 retained datasets are pooled with Rubin variance and finite-sample degrees of freedom. Both assume missingness can be explained by the included observed information. The panels target the same direction of change but rely on different populations or missingness assumptions. Ratios below one indicate lower W24 values; the line at one denotes no change. Bars are 95% confidence intervals. No observed value was replaced. Sparse donor support makes the abro 200 mg estimate sensitive to the imputation method; neither imputation confirms its direction of change. CAMS is used because Huashan has no follow-up IgE for dupi. Full-cohort, other-window and missing-not-at-random analyses are retained in Supplementary Table S8 and source data. No W4 IgE was observed, so no W4 point is imputed or plotted. The recorded value 5000 was not assumed to be a verified censoring threshold. These observational analyses do not identify causal drug effects.

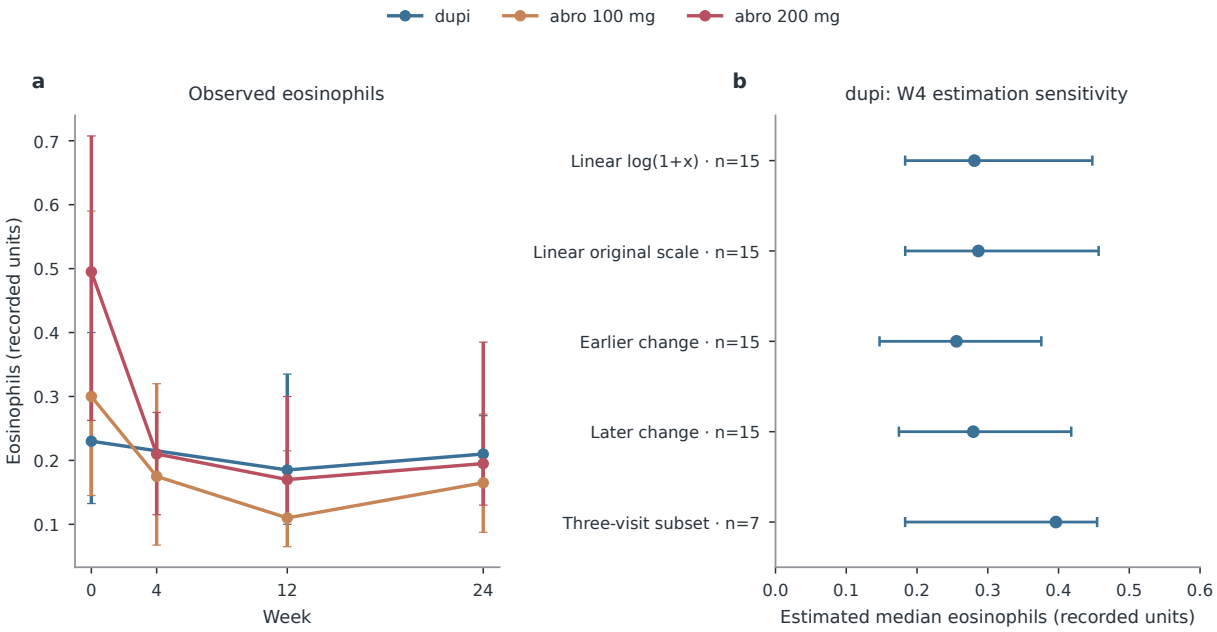


Figure S6: **Observed eosinophil data and sensitivity of the W4 estimate.** a, Available-case medians and interquartile ranges at observed visits. The dupi line connects W0 to W12 because no W4 measurement exists; crossing the W4 position does not create an observation. No imputed value enters this panel. b, Estimated dupi W4 medians and patient-bootstrap 95% intervals under different temporal assumptions. The first four estimates use the same 15 W0/W12 pairs. Earlier/later change replaces the linear time fraction with $\log(1+t)/\log(13)$ or $(t/12)^2$ at t=4. The last row restricts to seven patients also observed at W24, illustrating selection sensitivity, not a better estimate. Intervals condition on the stated assumptions and do not cover all uncertainty about the unmeasured W4 distribution. These are post hoc descriptive estimates, not multiple imputation under an identified treatment-specific W4 model. All original observations remain unchanged.

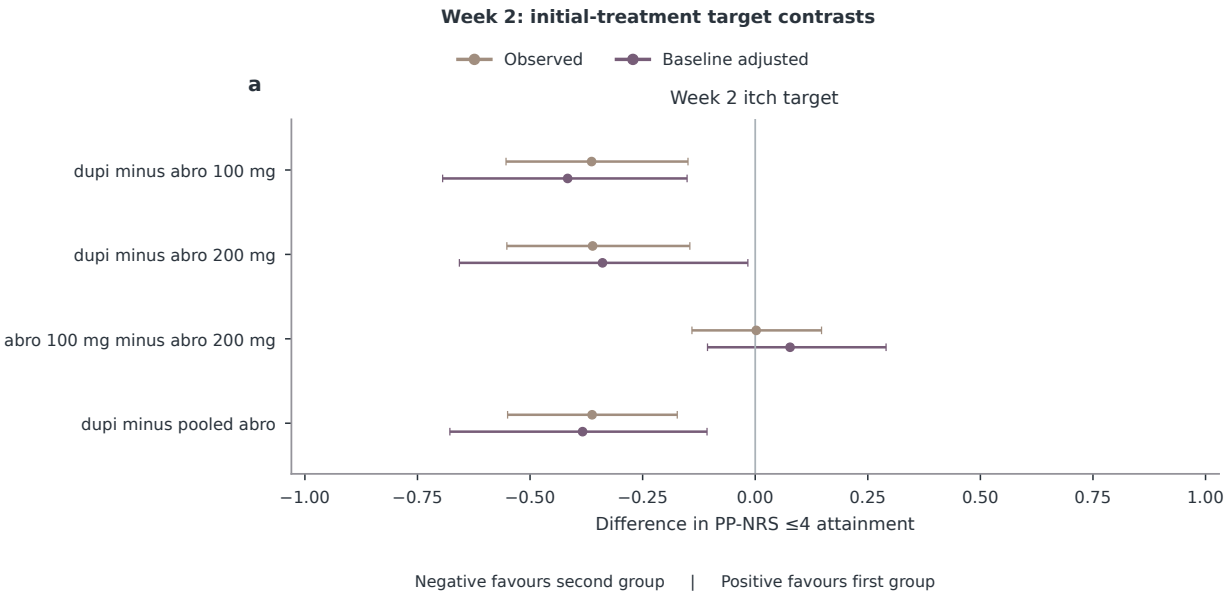


Figure S7: **Week 2: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

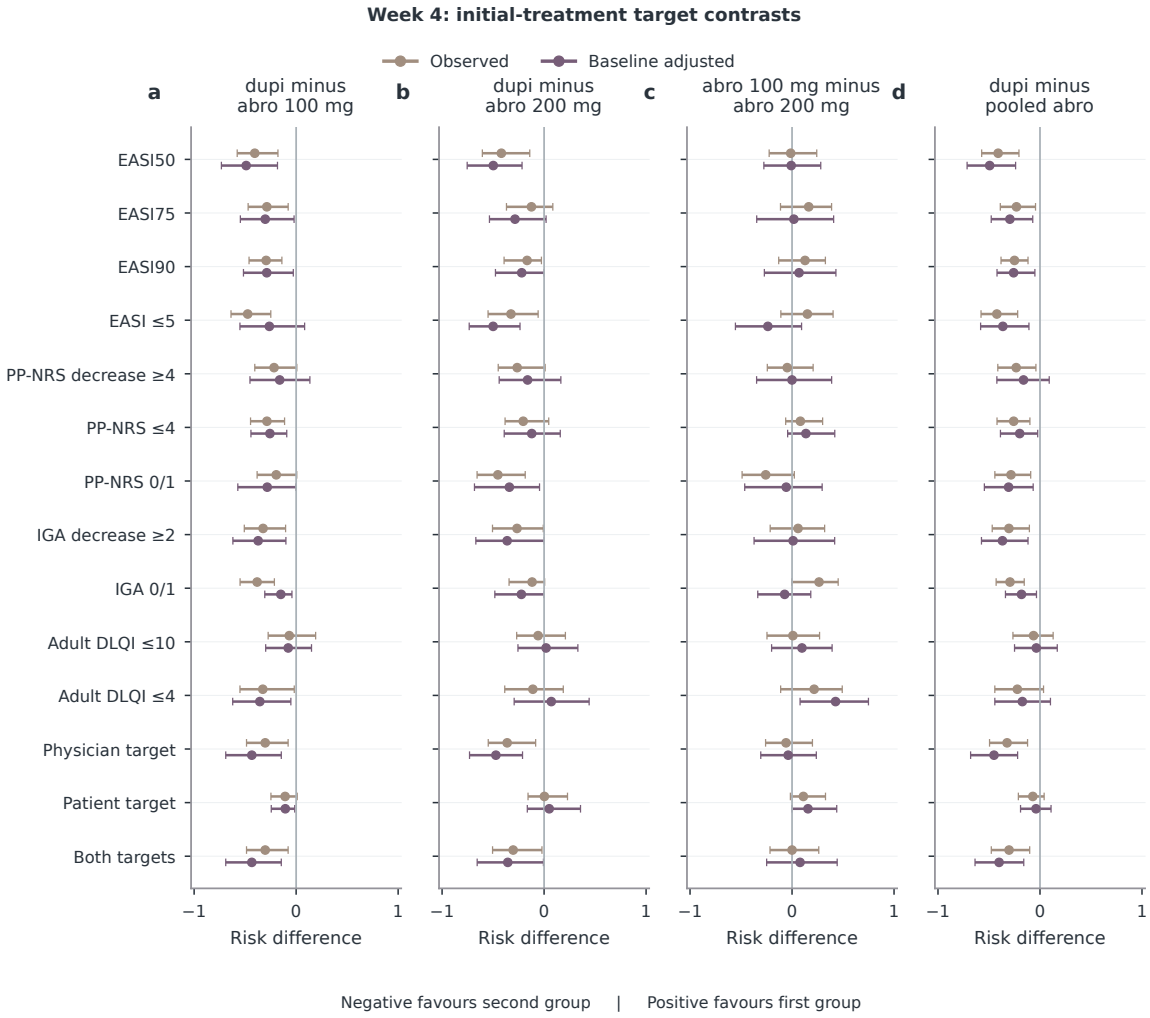


Figure S8: **Week 4: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

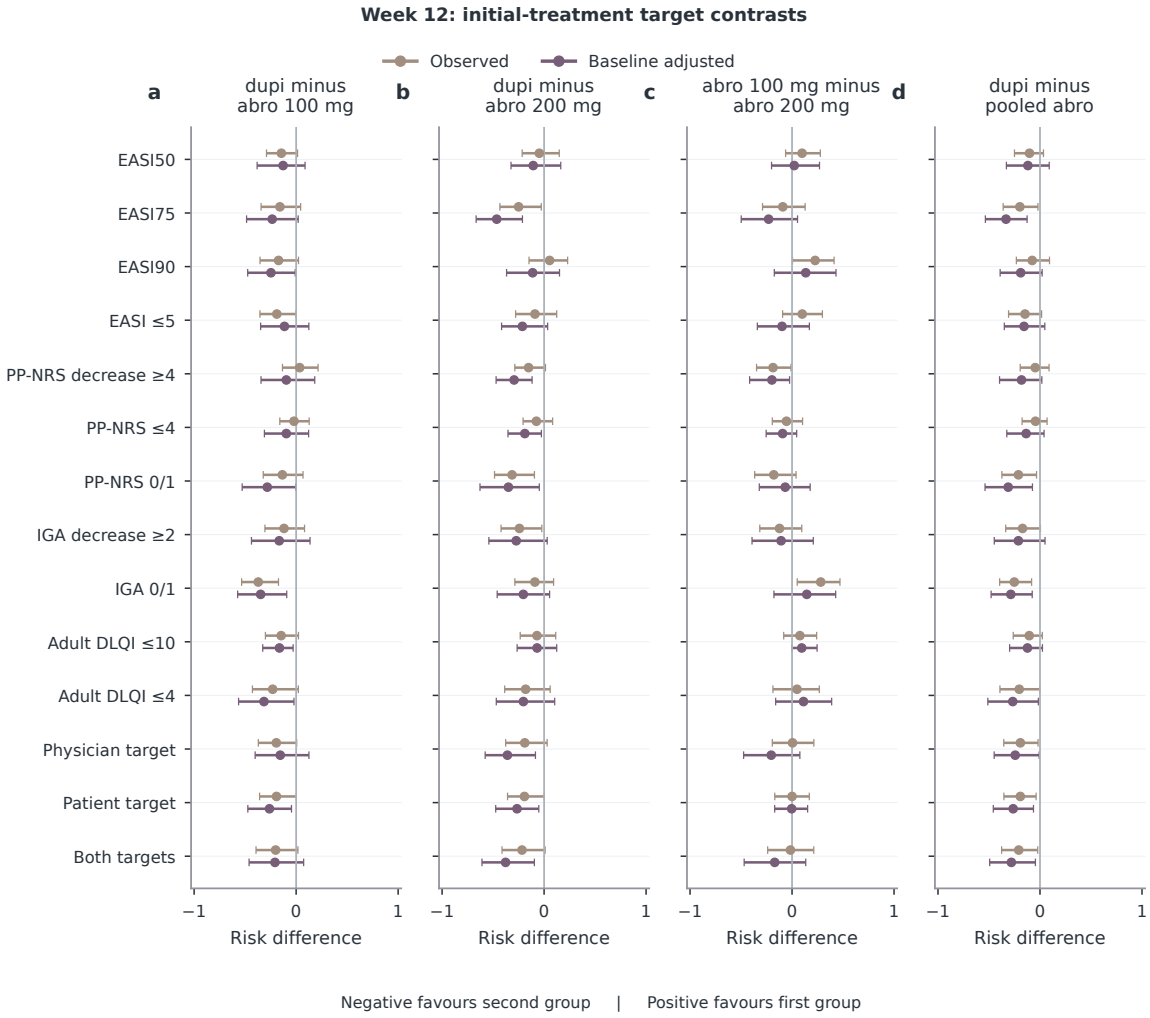


Figure S9: **Week 12: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

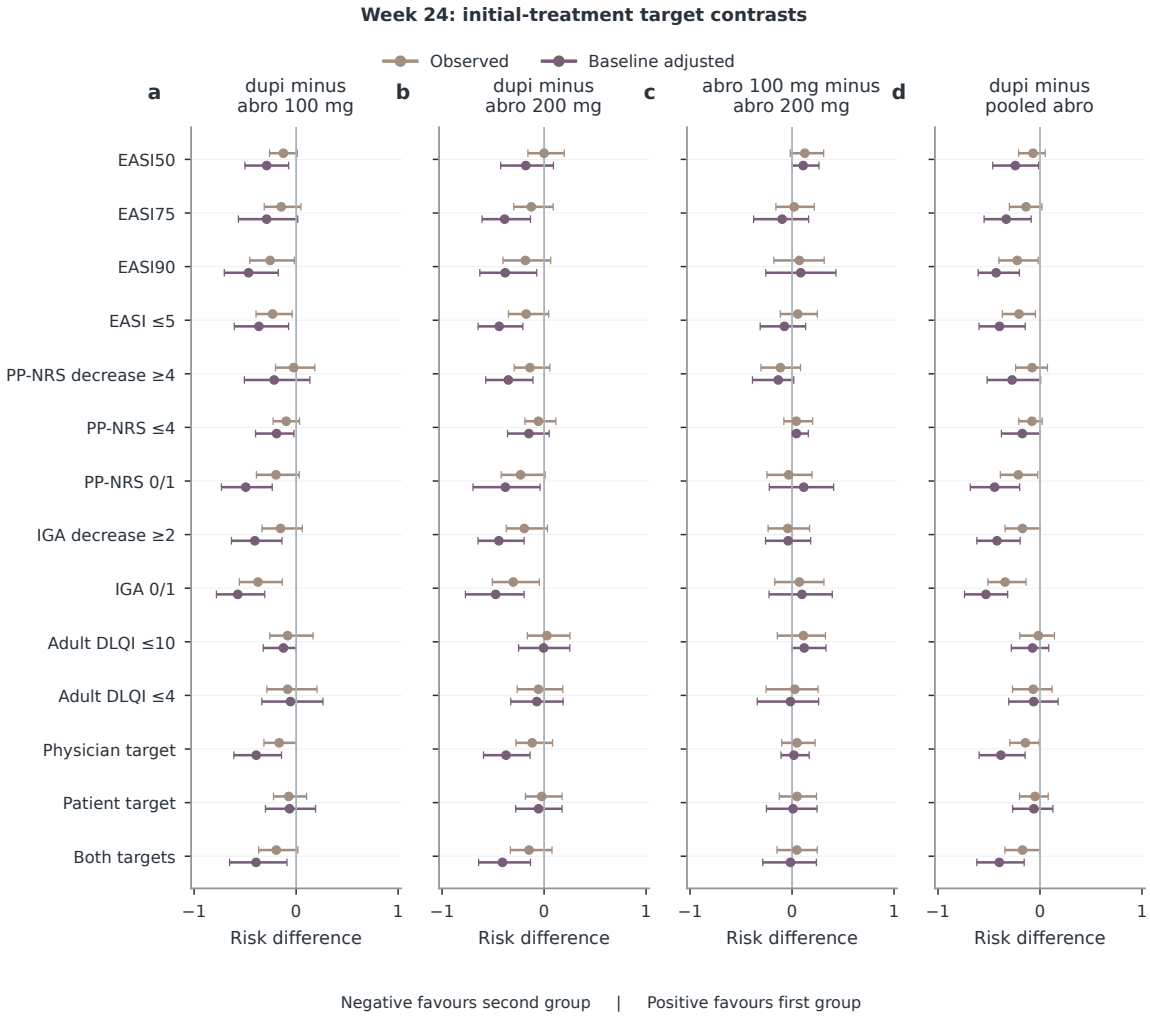


Figure S10: **Week 24: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

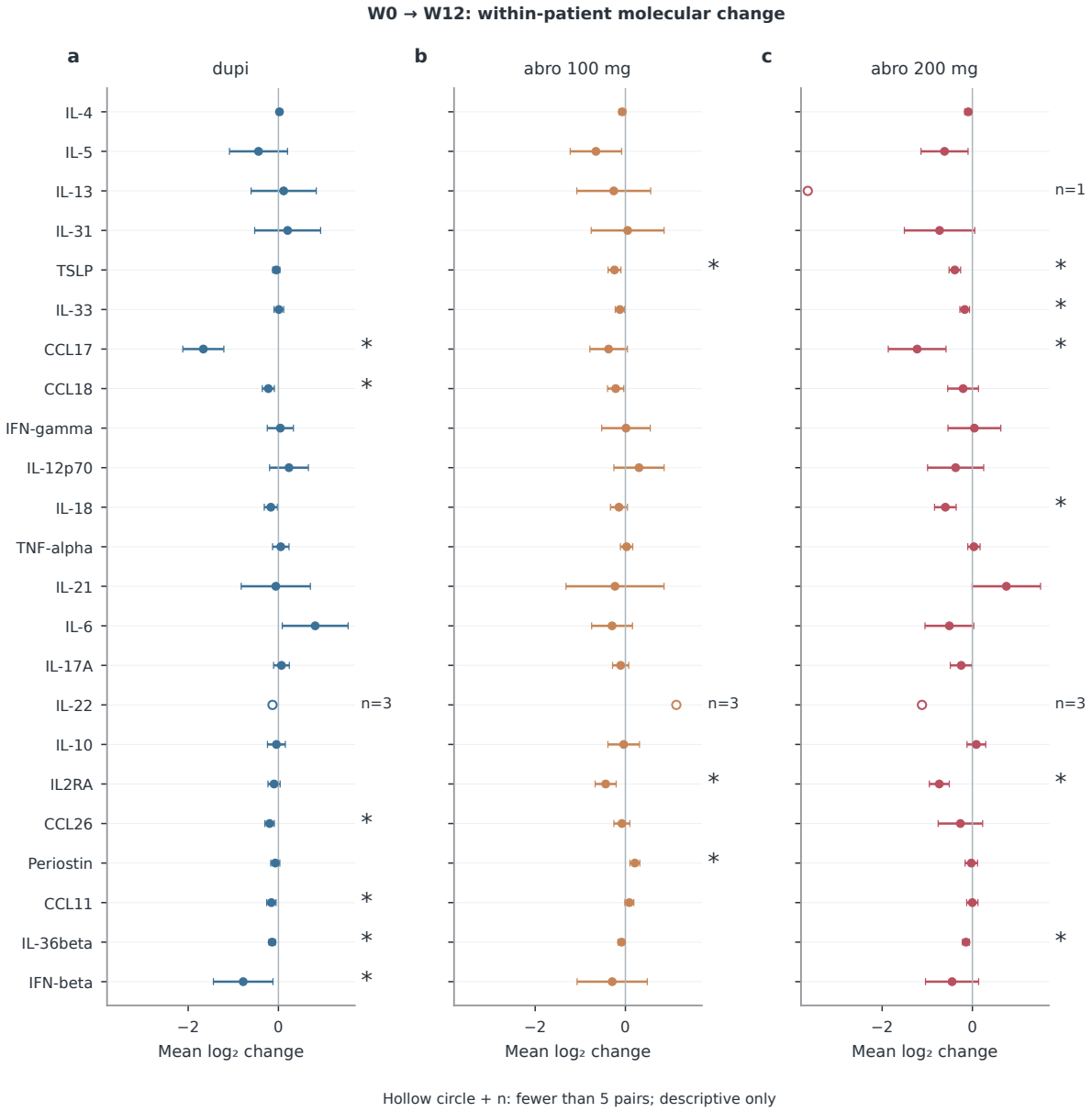


Figure S11: **Molecular changes from W0 to W12.** Mean paired $\log_2$ concentration changes with 95% t confidence intervals. Negative values indicate lower later concentration. Each marker/treatment uses all available pairs for this interval; exact counts are in the source table. All 23 measured markers are retained. Hollow circles with n labels denote fewer than five pairs; these are descriptive and have no inferential interval or star. Unavailable estimates are blank. Stars denote BH q<0.05 across the 193 estimable tests among 207 planned marker-by-treatment-by-window comparisons, not a separate family for the displayed interval.

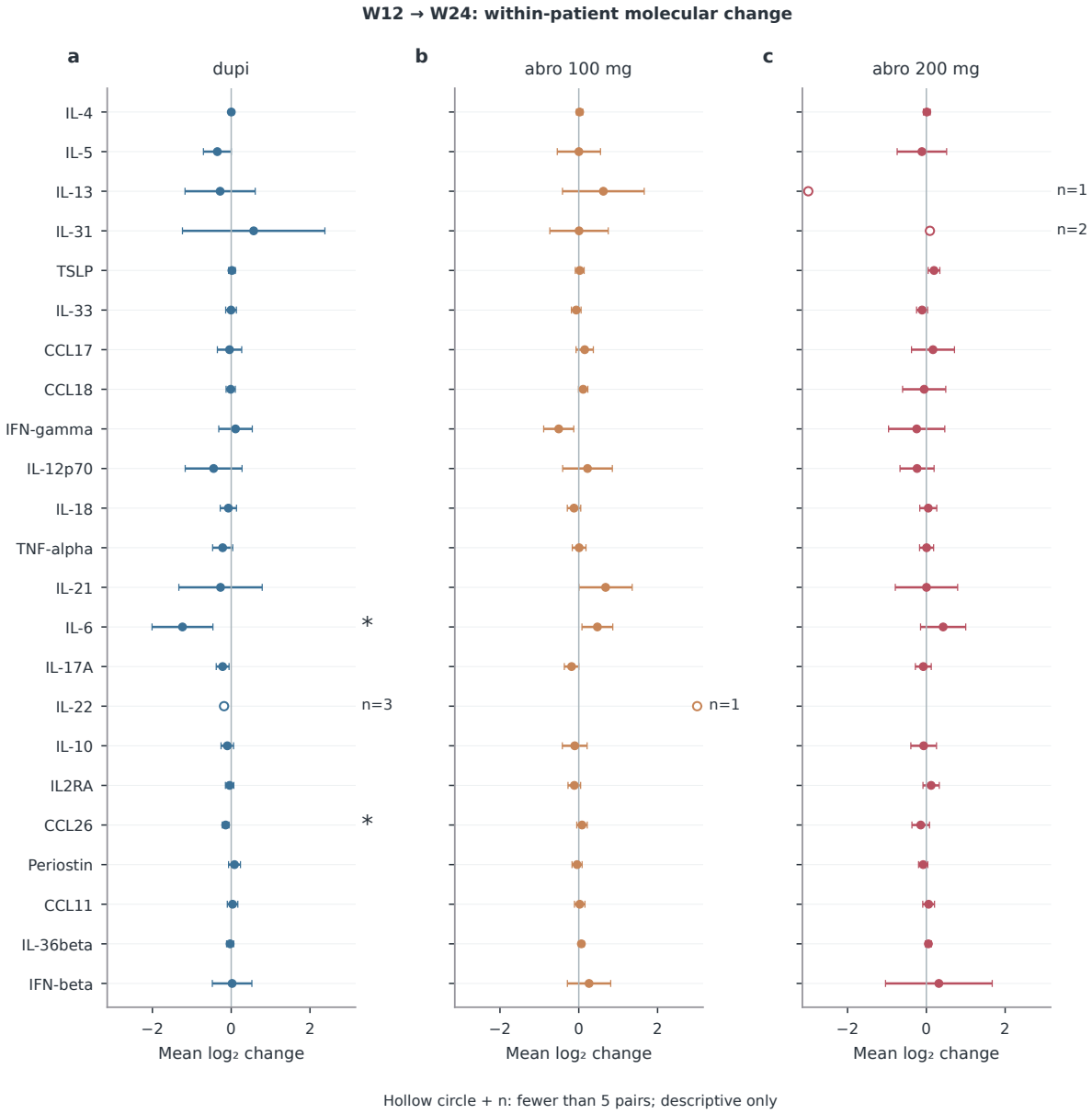


Figure S12: **Molecular changes from W12 to W24.** Mean paired $\log_2$ concentration changes with 95% t confidence intervals. Negative values indicate lower later concentration. Each marker/treatment uses all available pairs for this interval; exact counts are in the source table. All 23 measured markers are retained. Hollow circles with n labels denote fewer than five pairs; these are descriptive and have no inferential interval or star. Unavailable estimates are blank. Stars denote BH $q<0.05$ across the 193 estimable tests among 207 planned marker-by-treatment-by-window comparisons, not a separate family for the displayed interval.

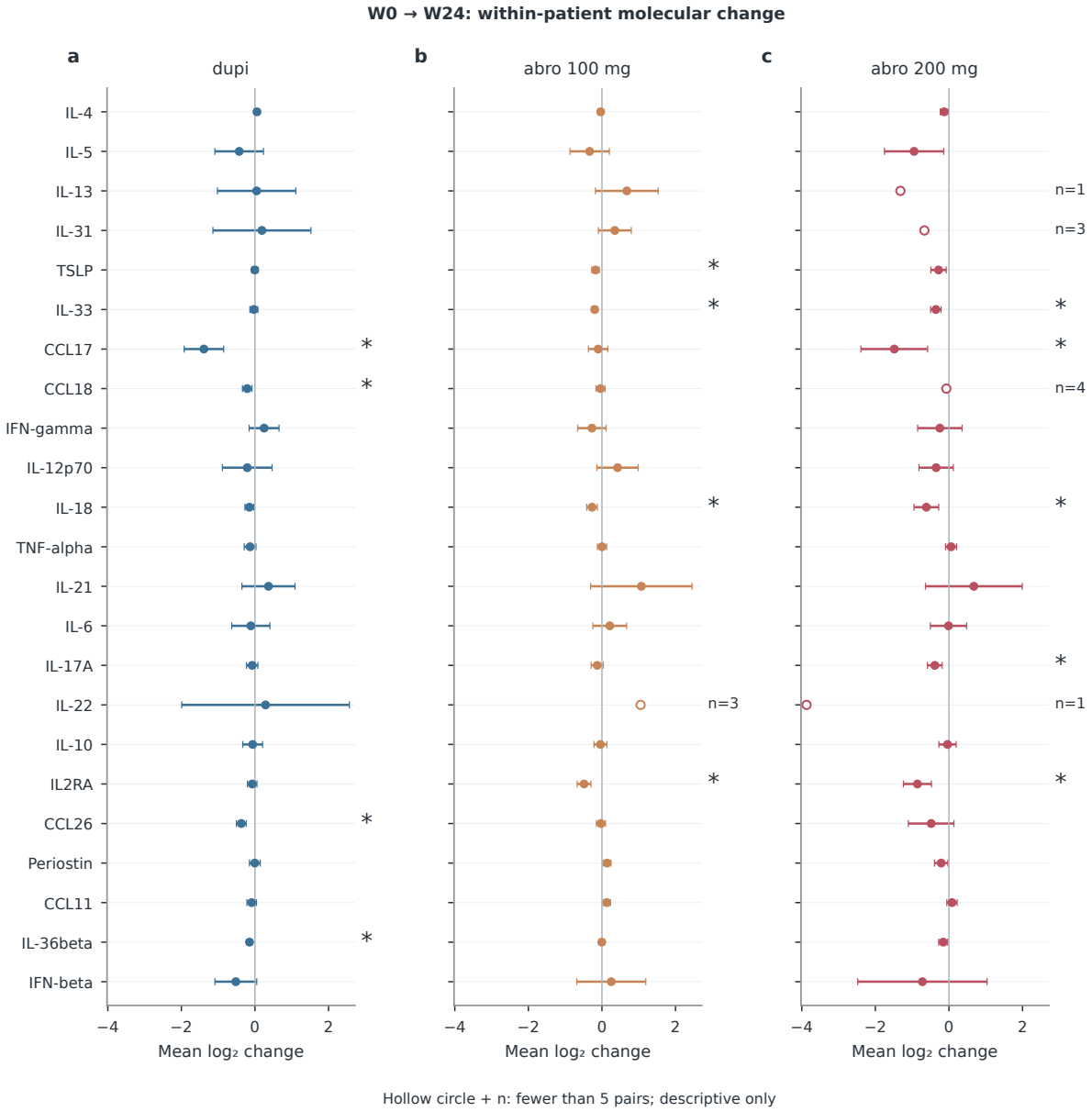


Figure S13: **Molecular changes from W0 to W24.** Mean paired $\log_2$ concentration changes with 95% t confidence intervals. Negative values indicate lower later concentration. Each marker/treatment uses all available pairs for this interval; exact counts are in the source table. All 23 measured markers are retained. Hollow circles with n labels denote fewer than five pairs; these are descriptive and have no inferential interval or star. Unavailable estimates are blank. Stars denote BH $q<0.05$ across the 193 estimable tests among 207 planned marker-by-treatment-by-window comparisons, not a separate family for the displayed interval.

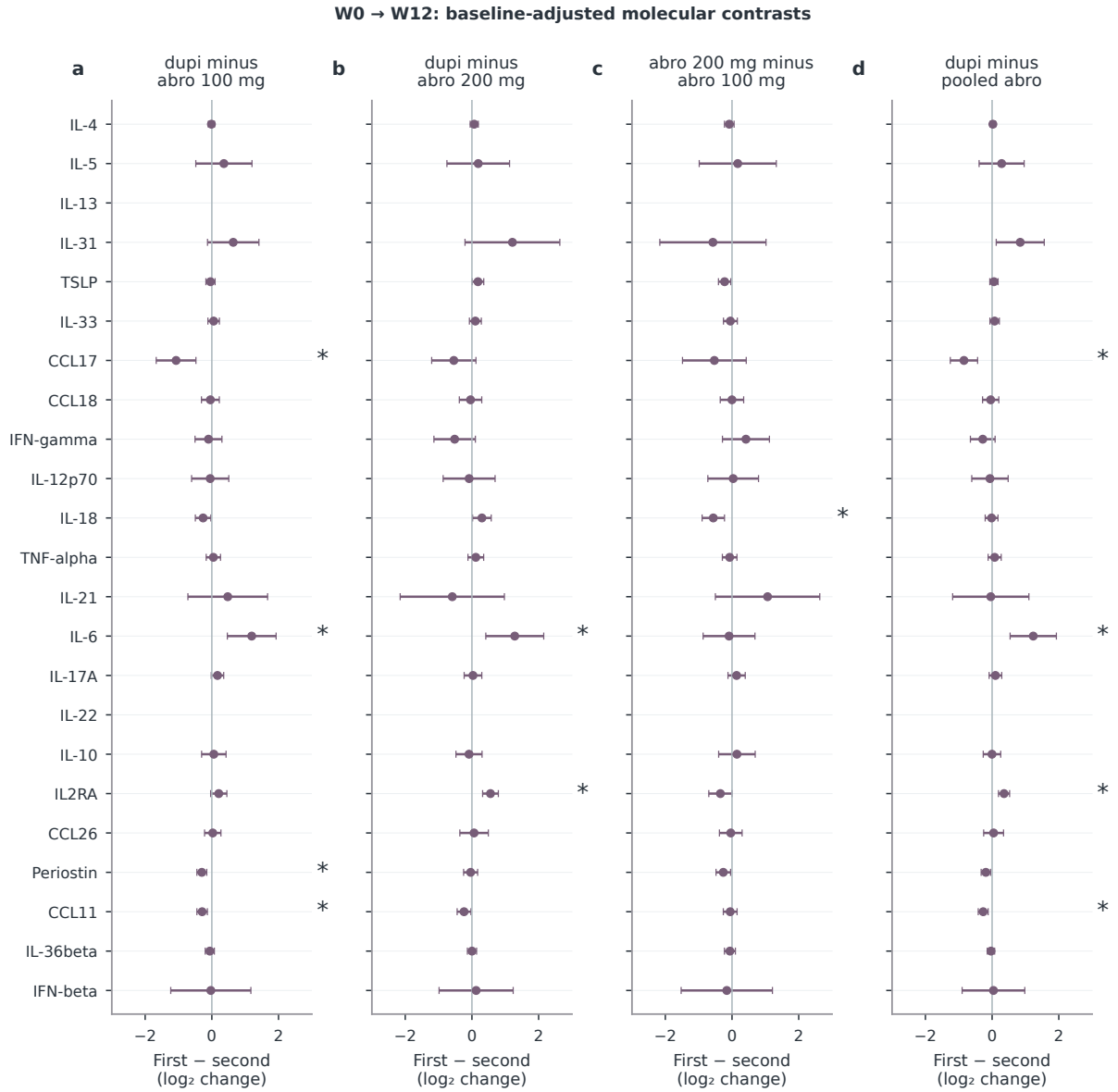


Figure S14: **Adjusted molecular comparisons from W0 to W12.** Differences in within-patient $\log_2$ changes with HC3 95% confidence intervals. Negative values mean a smaller, more negative change in the first named treatment. Models adjust starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre. dupi denotes dupilumab; abro denotes abrocitinib, with 100/200 mg identifying initial dose. pooled abro retains the observed dose mixture. Stars denote the original BH q<0.05 family; no multiplicity family is recalculated after display selection. Blank positions have no estimable contrast. Complete counts, diagnostics, centre/batch sensitivities and tests remain in the source table. These are adjusted observational associations, not randomized current-dose effects.

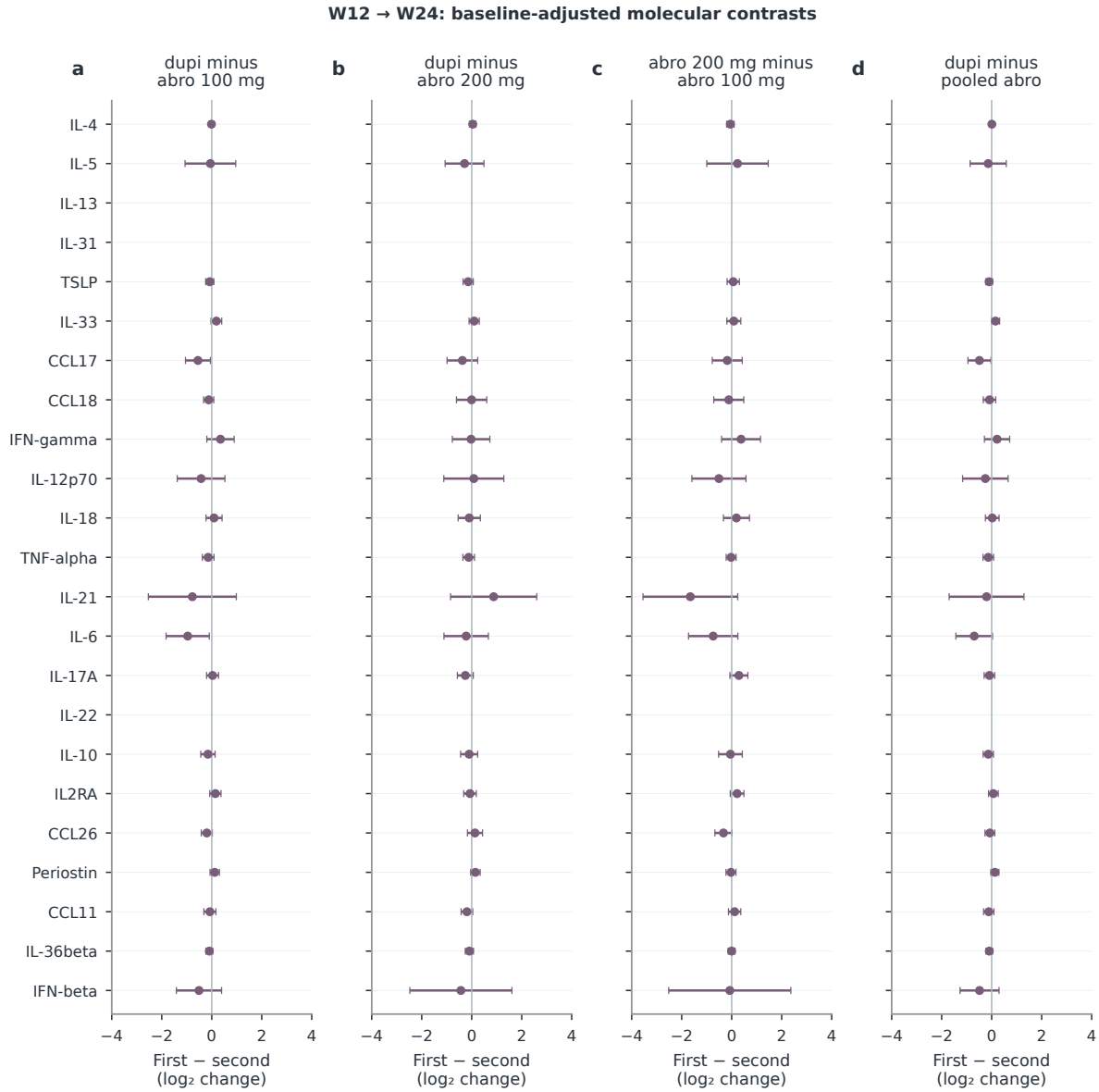


Figure S15: **Adjusted molecular comparisons from W12 to W24.** Differences in within-patient $\log_2$ changes with HC3 95% confidence intervals. Negative values mean a smaller, more negative change in the first named treatment. Models adjust starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre. dupi denotes dupilumab; abro denotes abrocitinib, with 100/200 mg identifying initial dose. pooled abro retains the observed dose mixture. Stars denote the original BH $q<0.05$ family; no multiplicity family is recalculated after display selection. Blank positions have no estimable contrast. Complete counts, diagnostics, centre/batch sensitivities and tests remain in the source table. These are adjusted observational associations, not randomized current-dose effects.

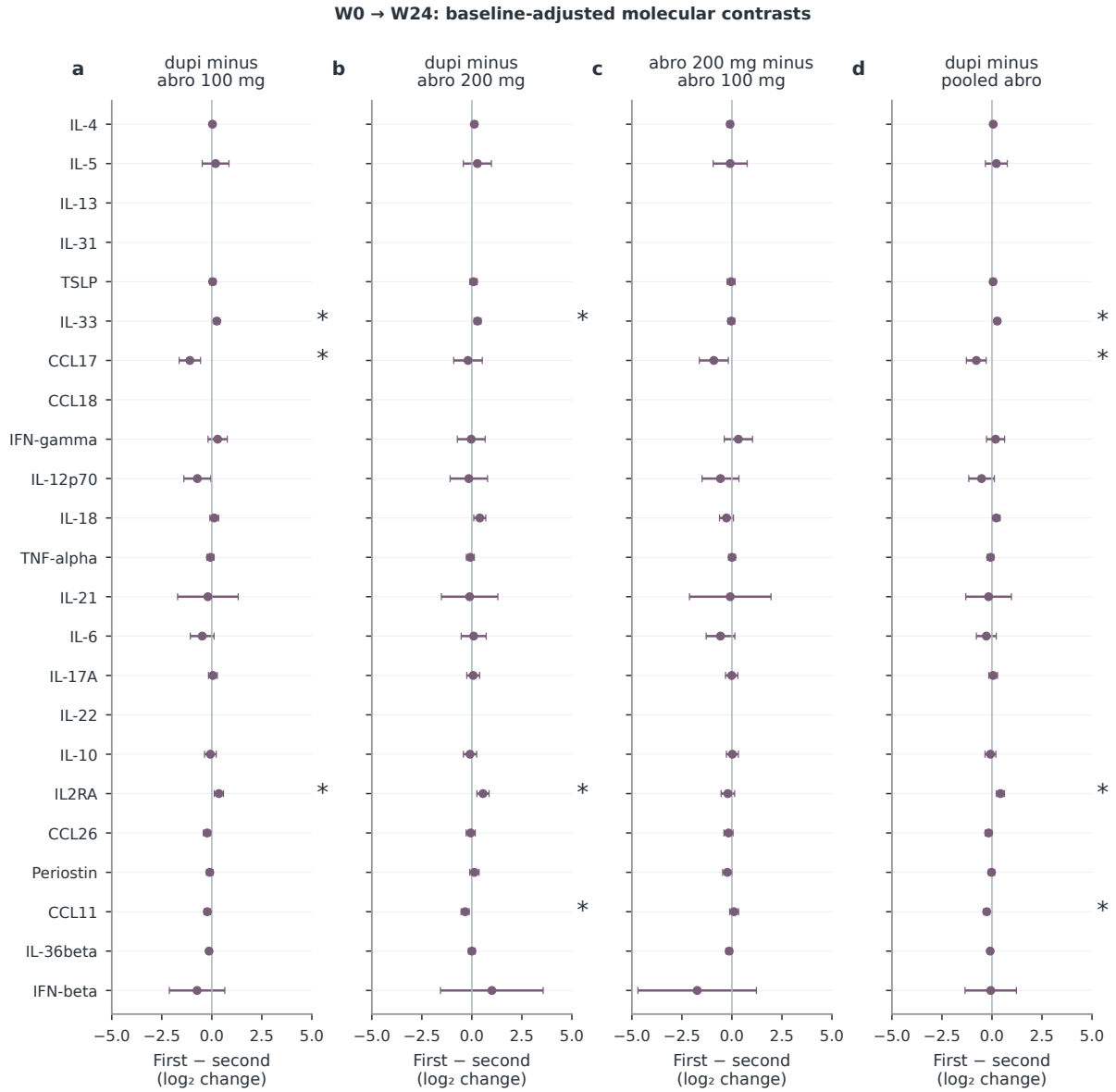


Figure S16: **Adjusted molecular comparisons from W0 to W24.** Differences in within-patient $\log_2$ changes with HC3 95% confidence intervals. Negative values mean a smaller, more negative change in the first named treatment. Models adjust starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre. dupi denotes dupilumab; abro denotes abrocitinib, with 100/200 mg identifying initial dose. pooled abro retains the observed dose mixture. Stars denote the original BH q<0.05 family; no multiplicity family is recalculated after display selection. Blank positions have no estimable contrast. Complete counts, diagnostics, centre/batch sensitivities and tests remain in the source table. These are adjusted observational associations, not randomized current-dose effects.

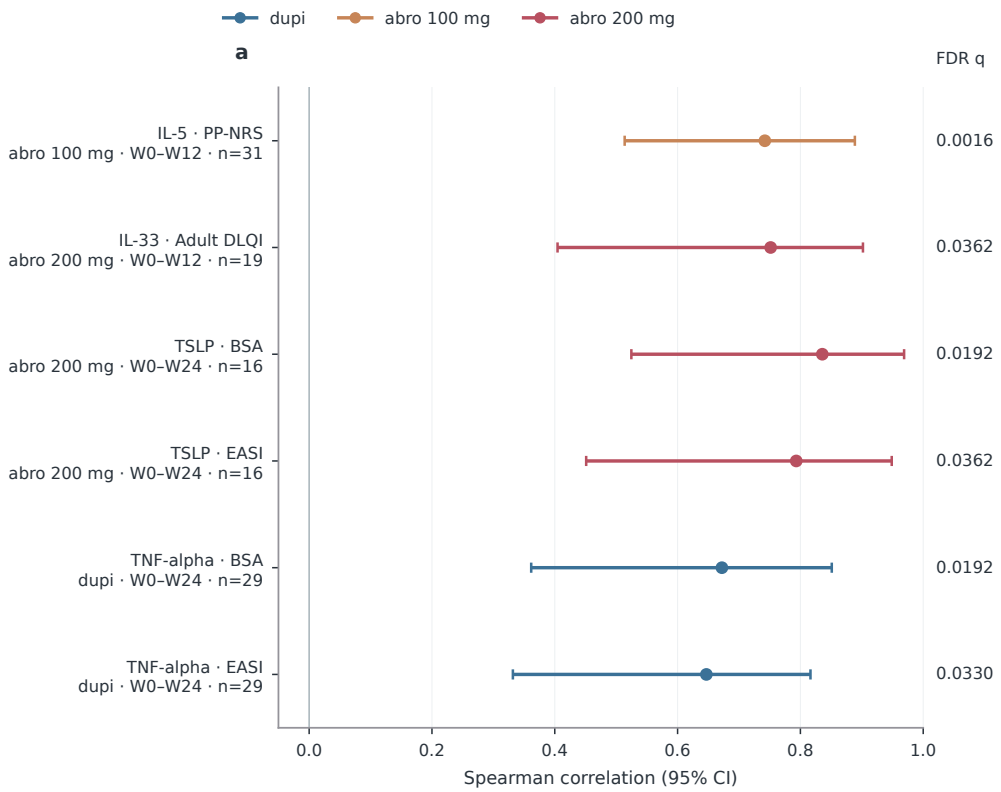


Figure S17: **Molecular and clinical changes associated after multiplicity correction.** Every association meeting BH q<0.05 in the full declared change-association family is displayed; selection spans 23 markers, five clinical outcomes, three initial treatments and three intervals. No W12–W24 association met this threshold. Points are Spearman correlations between within-patient clinical change and $\log_2$ molecular change, both later minus earlier, with patient-bootstrap 95% confidence intervals. Positive correlations indicate changes in the same direction. Labels give paired-patient counts and the interval; colours indicate initial treatment only. Adult DLQI requires age ≥18. All tested coefficients, non-significant results and unestimable pairs remain in the complete source table. This significance-selected display is exploratory and is not independent validation, numerical agreement or a causal molecular effect.

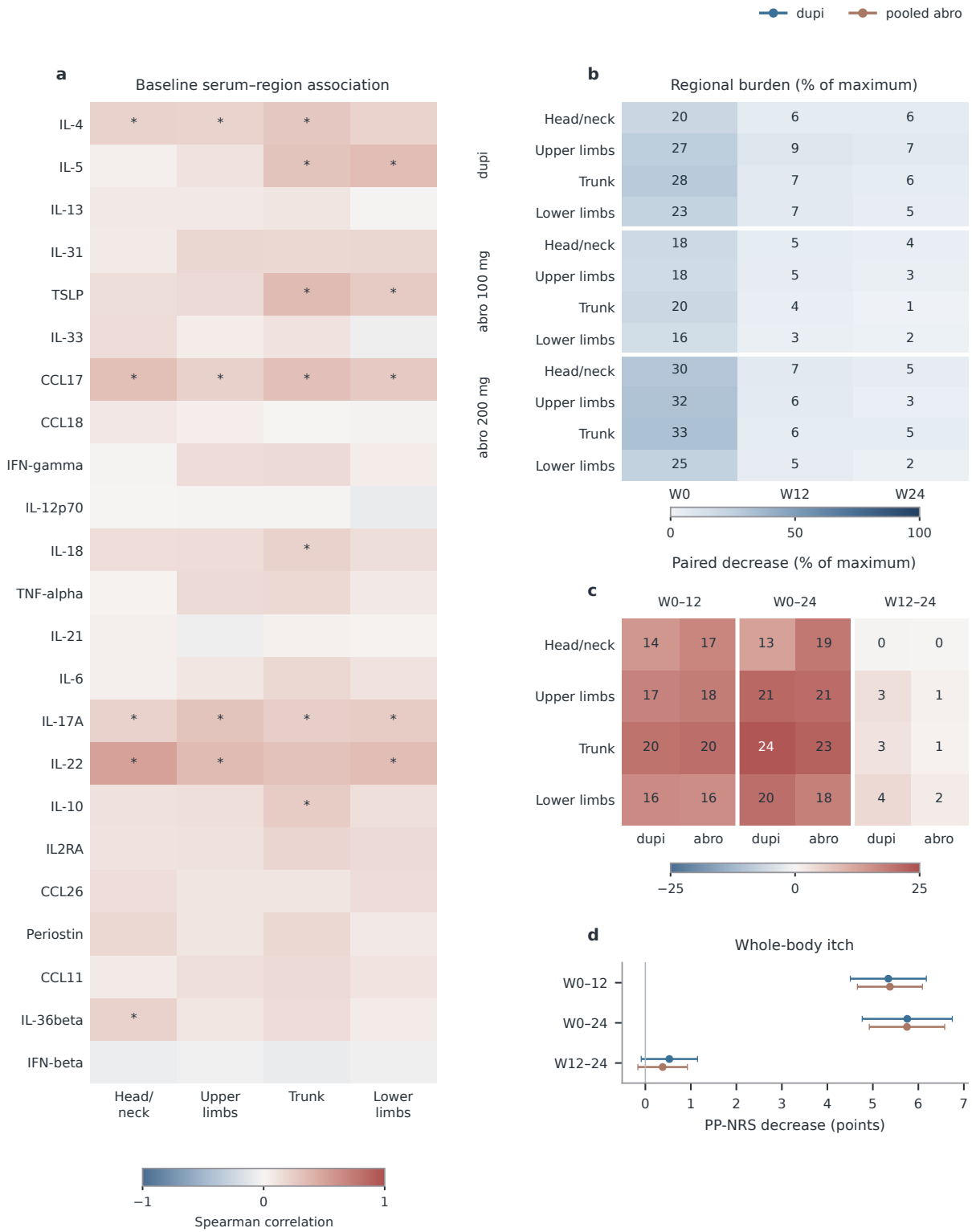


Figure S18: **Regional clinical burden and molecular associations.** a, Baseline Spearman associations between all 23 markers and the four regional EASI contributions; stars mark BH q<0.05, with complete coefficients, patient counts and bootstrap intervals retained in the source. b, Available-case mean regional EASI divided by its theoretical maximum and multiplied by 100; cells show percentages, not patient counts. c, Mean paired regional decrease as a percentage of the same regional maximum for three separately labelled intervals; positive values indicate improvement. The abro columns represent the observed pooled-dose mixture. These heatmaps summarize magnitude; exact 95% t intervals and counts are in the source tables. d, Whole-body PP-NRS mean decreases with 95% t intervals, not region-specific itch. Regional maxima are 7.2, 14.4, 21.6 and 28.8 for head/neck, upper limbs, trunk and lower limbs. All regional analyses require complete, bounded regional records consistent with total EASI. Intervals can share patients and are not successive cumulative effects. Colour bars encode distinct quantities and are not interchangeable. No baseline-adjusted regional treatment contrast passed the FDR threshold; full separate-dose comparisons remain available.

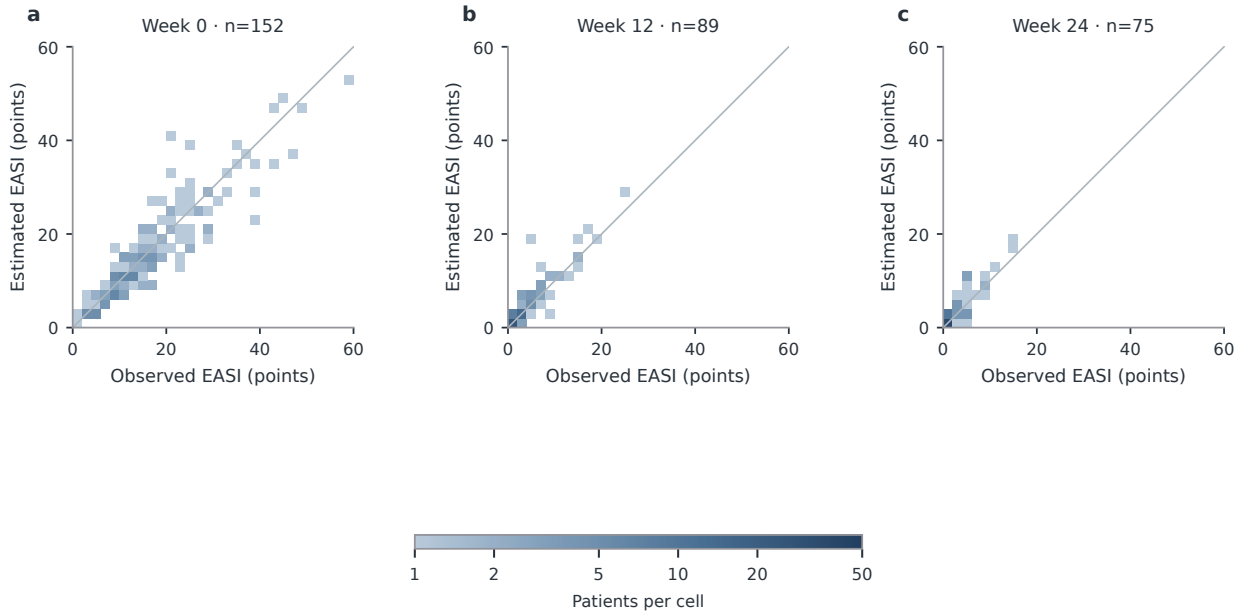


Figure S19: **Observed and estimated EASI at each visit.** a–c, Estimated versus observed EASI in the first patient-held-out repetition at W0/W12 and same-cohort temporal application at W24. Each fixed 2×2-EASI-point cell counts all exact score pairs within its boundaries. Colour gives patient count on one shared logarithmic scale; white cells contain no observations. All 152/89/75 eligible patients are retained. Identical axes preserve the severity-range comparison; grey diagonals denote exact agreement. W0/W12 use nested patient-level structure and penalty selection; earlier marker selection is not repeated. W24 had previously been inspected. These results assess the development procedure rather than independent validation of the final coefficient vector. Overall performance is in Table 2; the main figure displays the complete absolute-error distribution.

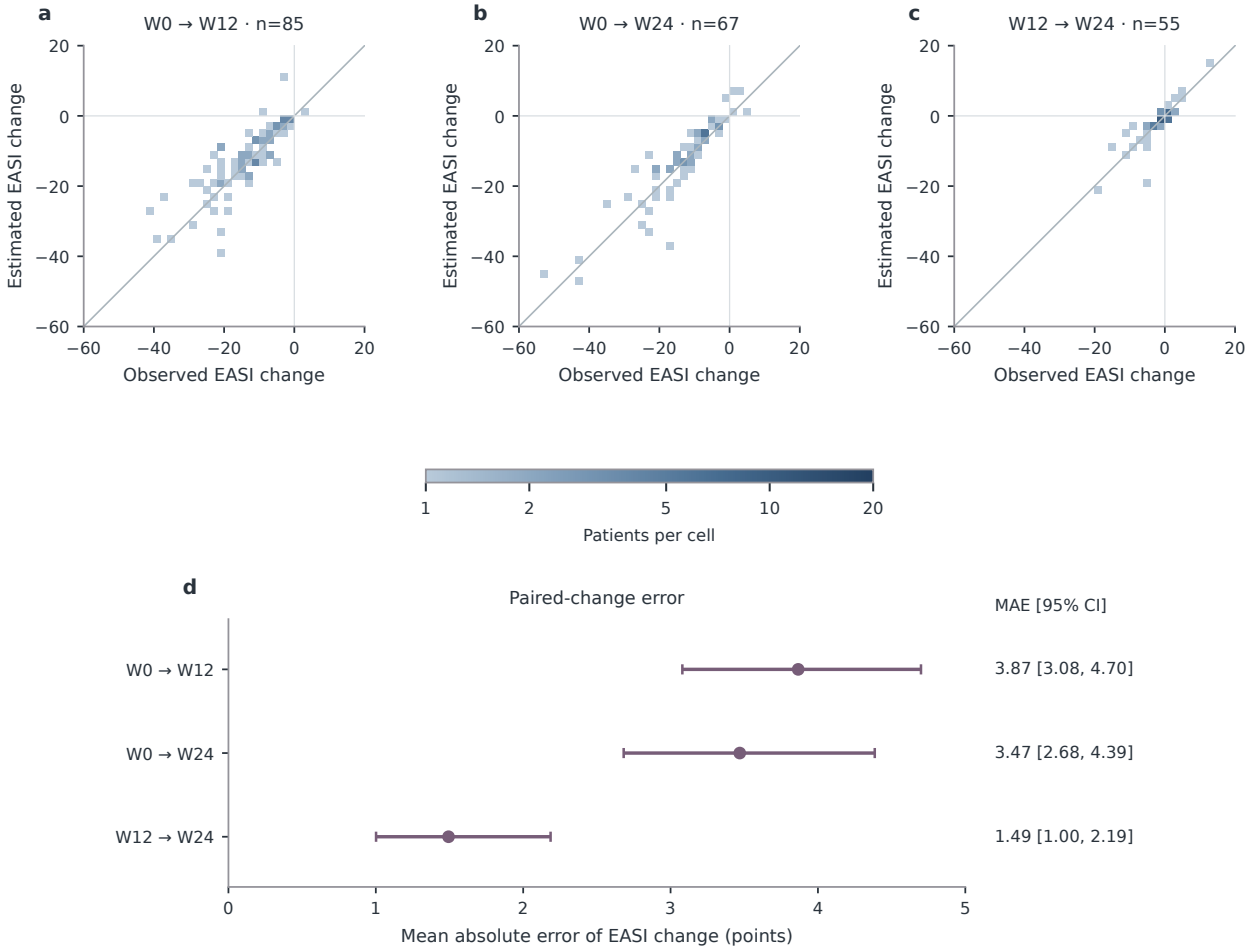


Figure S20: **Within-patient EASI change and absolute error.** a–c, Estimated versus observed EASI changes in the first patient-held-out repetition. Change is end minus start: negative values indicate improvement. Fixed 2×2-EASI-point cells count exact paired changes on one logarithmic count scale. Identical axes retain all 85, 67 and 55 pairs; the intervals can share patients and are not consecutive categories on a time axis. White cells contain no observations; diagonals mark exact agreement. d, Change MAE calculated within each of two repetitions before averaging, with 2,000 patient-bootstrap 95% confidence intervals conditional on saved predictions. Both scores are paired within each repetition before subtraction. W24 is previously inspected same-cohort temporal application. EASI75 classification is shown in the main model figure; complete EASI50/75/90 classifications are retained separately.

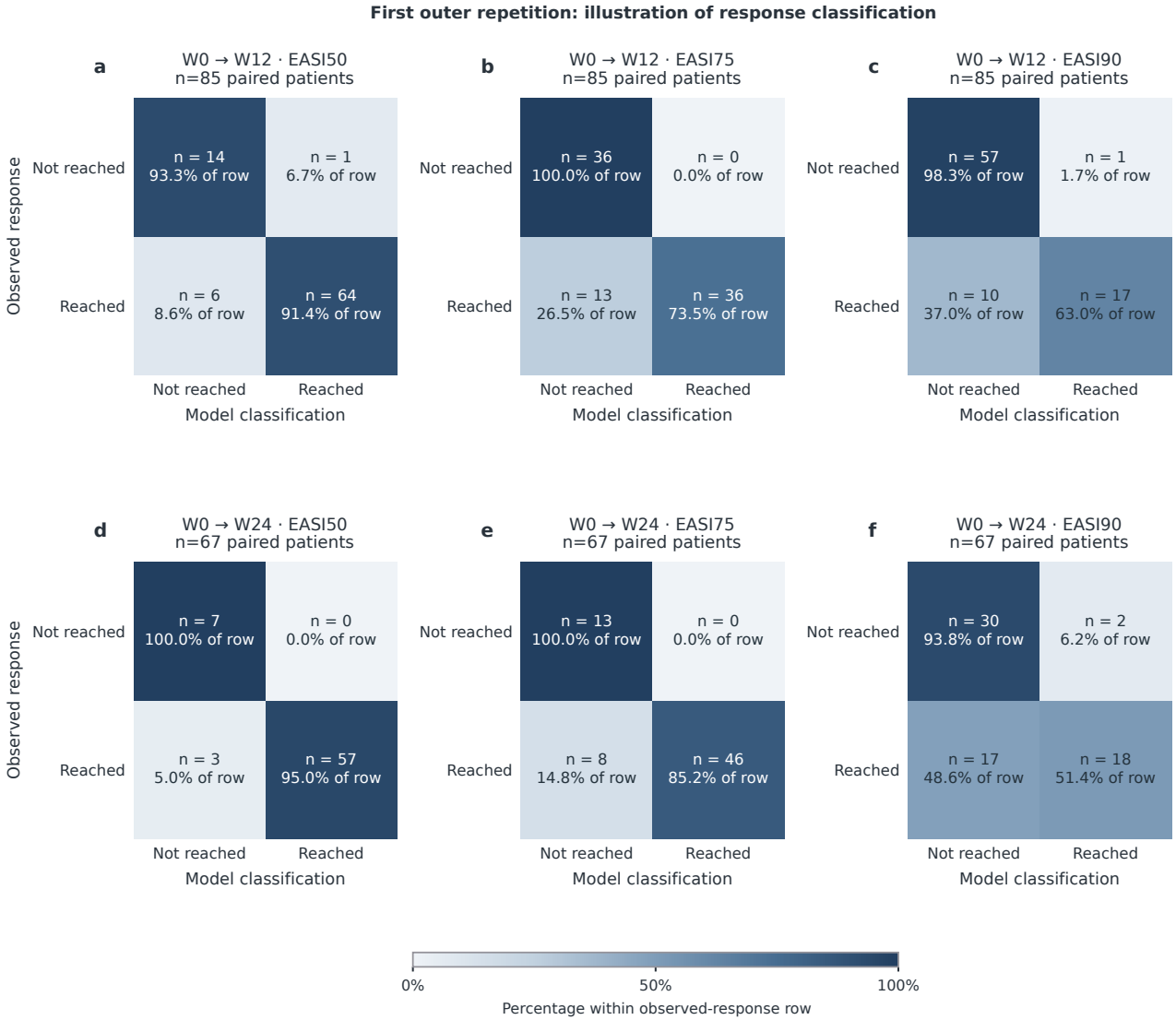


Figure S21: **Classification of EASI50, EASI75 and EASI90 response.** Each panel shows actual patient counts from the first outer repetition, separately for W0→W12 and W0→W24. Rows are observed response and columns model-estimated response. Diagonal cells are correct; upper-right cells falsely indicate response and lower-left cells miss an observed responder. Percentages are within each observed-response row, so the lower-right percentage is sensitivity and the upper-left percentage specificity. Observed response uses observed baseline/follow-up scores, whereas estimated response uses predicted baseline/follow-up scores. Counts are integers from one repetition, not the primary two-repetition average; the full metrics table retains the latter. The paired denominators are 85 and 67, with no excluded pair for invalid baseline in this displayed repetition. The two intervals can share patients. This assesses response classification, not alternative-treatment outcomes or continuous-score equivalence.